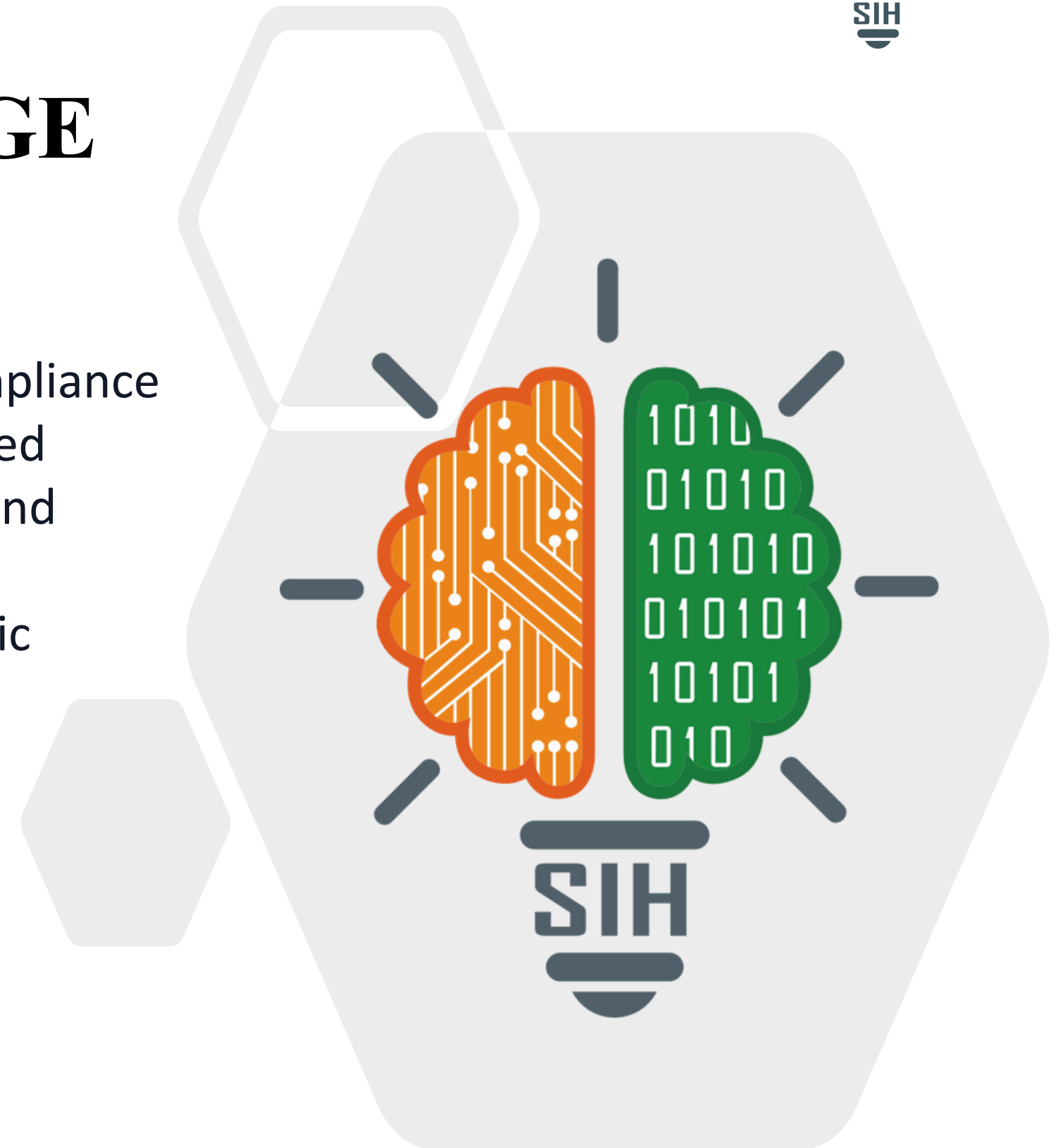


TITLE PAGE

- **Problem Statement ID** – 26034
- **Problem Statement Title** – Software System to check compliance of Packaged Commodities under Legal Metrology (Packaged Commodities) Rules, 2011 by scanning products, images and labels
- **Organization** – Ministry of Consumer Affairs, Food & Public Distribution
- **Department** – Department of Consumer Affairs (DoCA)
- **Theme** – Miscellaneous
- **PS Category** – Software
- **Team ID** – <Your Portal Team ID>
- **Team Name** – Metrolens



PRINCIPAL DISPLAY PANEL

SUNRISE

Refined Sunflower Oil

Net Qty. 500 g

M.R.P. ₹ 145.00

(incl. of all taxes)

Mfd. by Sunrise Foods Pvt. Ltd.



8 901234 567890

MEASURED PANEL AREA

182 cm²

RULE 7(2) MINIMUM

2.5 mm

2.5 mm

✓ Net quantity — compliant

2.0 mm

✗ M.R.P. — non-compliant

0.5 mm short of the minimum

Same-looking text. Different measured height.

✓ Packer name & address

✓ Net quantity

✓ Consumer care details

✗ Unit sale price

Rule 6 declarations audited in the same pass.

THE GAP TODAY

- Font size is a legal measurement, but field inspection is visual.
- A photograph has pixels, not millimetres.
- Borderline cases need evidence, not guesses.
- Manual inspection does not scale across products and listings.

HOW METROLENS WORKS

- Officer photographs the label with a reference card in frame.
- Perspective is corrected and the known card width converts pixels → millimetres.
- Declarations are extracted, validated and cross-checked, including unit-sale-price calculation.
- Rule + measurement + uncertainty produce the verdict and notice offline.

HOW IT ADDRESSES THE PROBLEM

- Font size stops being a judgement call and becomes a measurement with a stated error.
- Missing declarations, wrong placement and altered price stickers are caught in the same pass.
- Short weight is flagged and routed to a certified scale, never inferred from an image.

INNOVATION AND UNIQUENESS OF THE SOLUTION

MEASURES, NOT JUST READS

Returns physical numeral height in millimetres from a photograph.

KNOWS WHEN NOT TO DECIDE

If uncertainty crosses the legal limit, it returns CANNOT DETERMINE and routes the pack to a caliper.

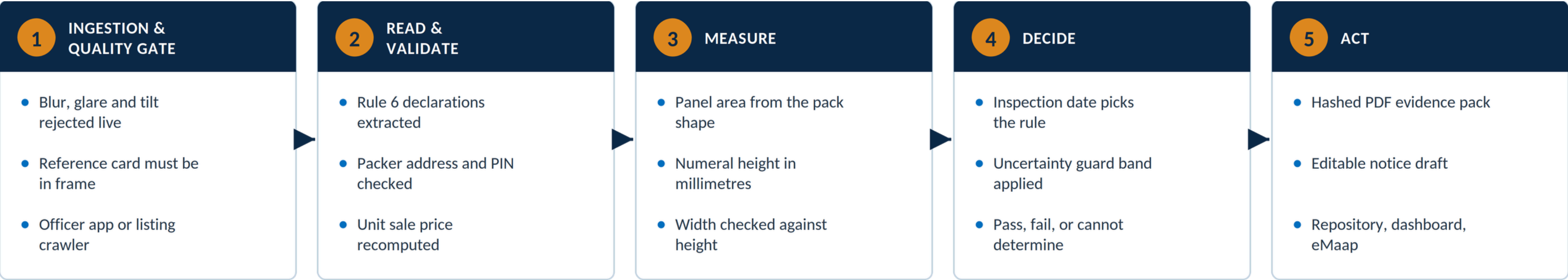
APPLIES THE RULE IN FORCE

Uses the applicable dated rule version rather than a static rulebook.

URNS FINDINGS INTO EVIDENCE

Produces a hashed evidence pack and editable notice offline.

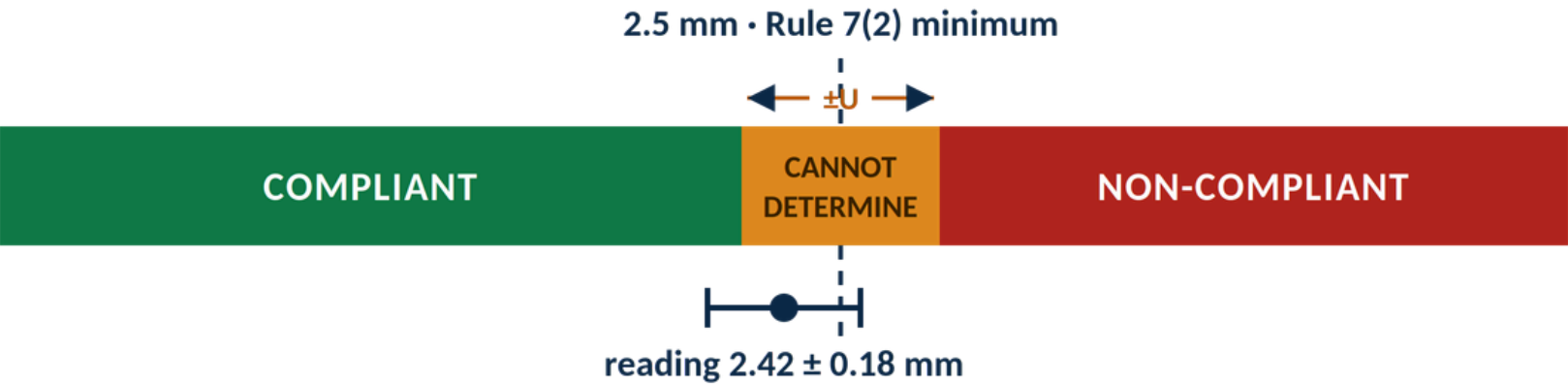
Five stages. The field path works offline.



TECHNOLOGY STACK

- **Field client** — Android and Kotlin, OpenCV NDK, offline SQLite store
- **Backend** — Python and FastAPI services, PostgreSQL repository
- **Vision and metrology** — OpenCV fiducial tracking and planar homography
- **OCR and extraction** — PP-OCR mobile with ML Kit, English and Devanagari
- **Dashboard** — React with three-tier role-based access control
- **Evidence integrity** — SHA-256 image hash, EXIF/GPS, reference ID and timestamp

UNCERTAINTY-AWARE DECISION



The interval crosses the limit, so no binary verdict is issued.
Guard banding per ILAC G8:09/2019. The pack goes to a caliper.

WHY A REFERENCE CARD

PHONE PHOTO ALONE



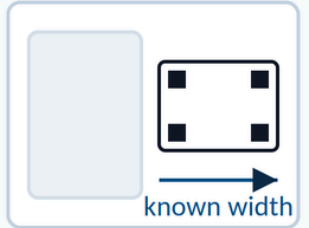
4032 × 3024 px
Scale: unknown
Physical mm: unknown

✗ Cannot determine compliance

A photograph alone carries no physical scale.

vs

PHOTO + REFERENCE CARD



- Corners to subpixel
- Perspective removed
- Pixels → millimetres

✓ Comparable to the legal limit

We put the scale into the frame, not into a guess.

ANALYSIS OF THE FEASIBILITY OF THE IDEA

- **Scale comes from the scene, not the sensor.** A printed fiducial in the same plane as the label carries it.
- **No capital expenditure.** It runs on the phones officers already carry.
- **Genuinely offline.** Reading, measurement and notice drafting all execute on-device.
- **Nothing new to integrate with.** eMaap, India's National Legal Metrology Portal, was launched on 24 December 2024.
- **No external dataset needed.** The caliper ground-truth corpus is built in-house.

CHALLENGES, RISKS AND HOW WE HANDLE THEM

Risk	Level	Mitigation
Blur or glare in the frame	HIGH	Rejected live, before any compute is spent
Curved or cylindrical packs	MED	Area falls back to the cylindrical formula in the Rules
OCR misreads stylised text	MED	Tap to select; the measurement never depends on OCR
Law amended mid-deployment	MED	Each rule version is mapped to its official effective date, and the engine applies the version valid on the inspection date.
Reading straddles the limit	HIGH	Guard banding returns indeterminate and calls for a caliper

The measurement is validated before the product is built. *If error exceeds 0.3 mm on known type sizes, we redesign rather than demo around it.*

POTENTIAL IMPACT ON THE TARGET AUDIENCE · BENEFITS OF THE SOLUTION

Visual judgement becomes measured evidence

LEGAL METROLOGY OFFICER

- A measured reading with a stated error replaces a judgement call
- The notice is drafted on the phone, at the shop
- Every finding carries its hash, location and rule

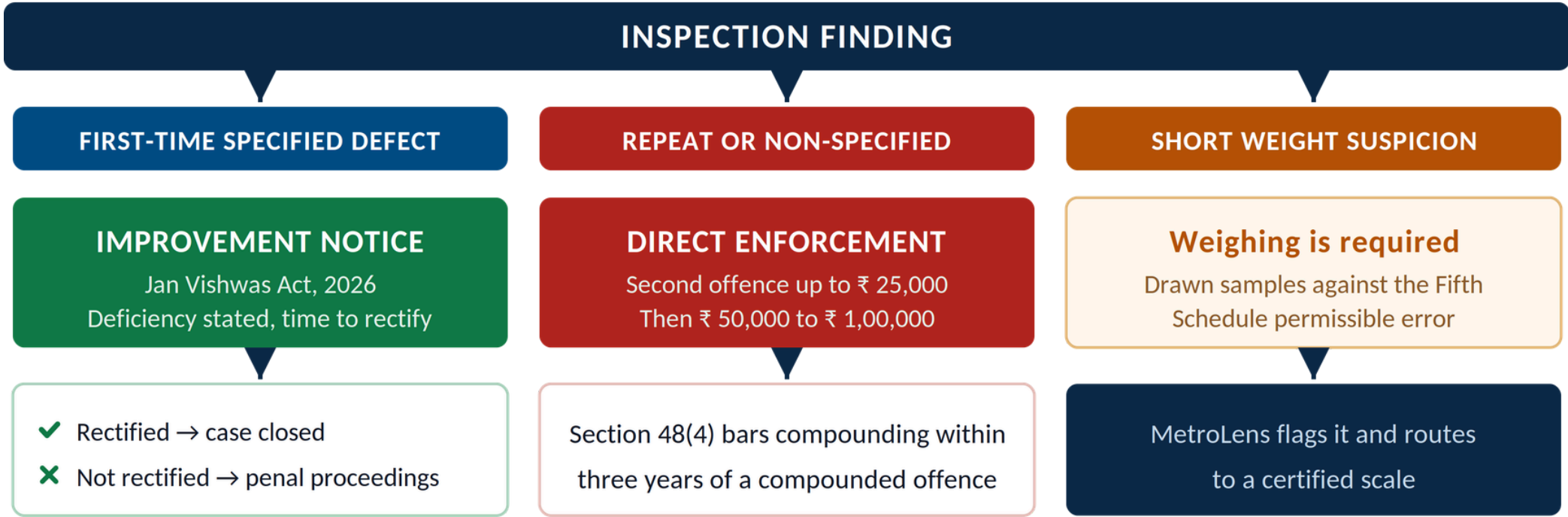
BUSINESS AND MSMEs

- A first specified defect gets an Improvement Notice, not a prosecution
- The notice names the exact deficiency and the time to fix it
- Honest packers stop losing shelf space to non-compliant rivals

CONSUMER

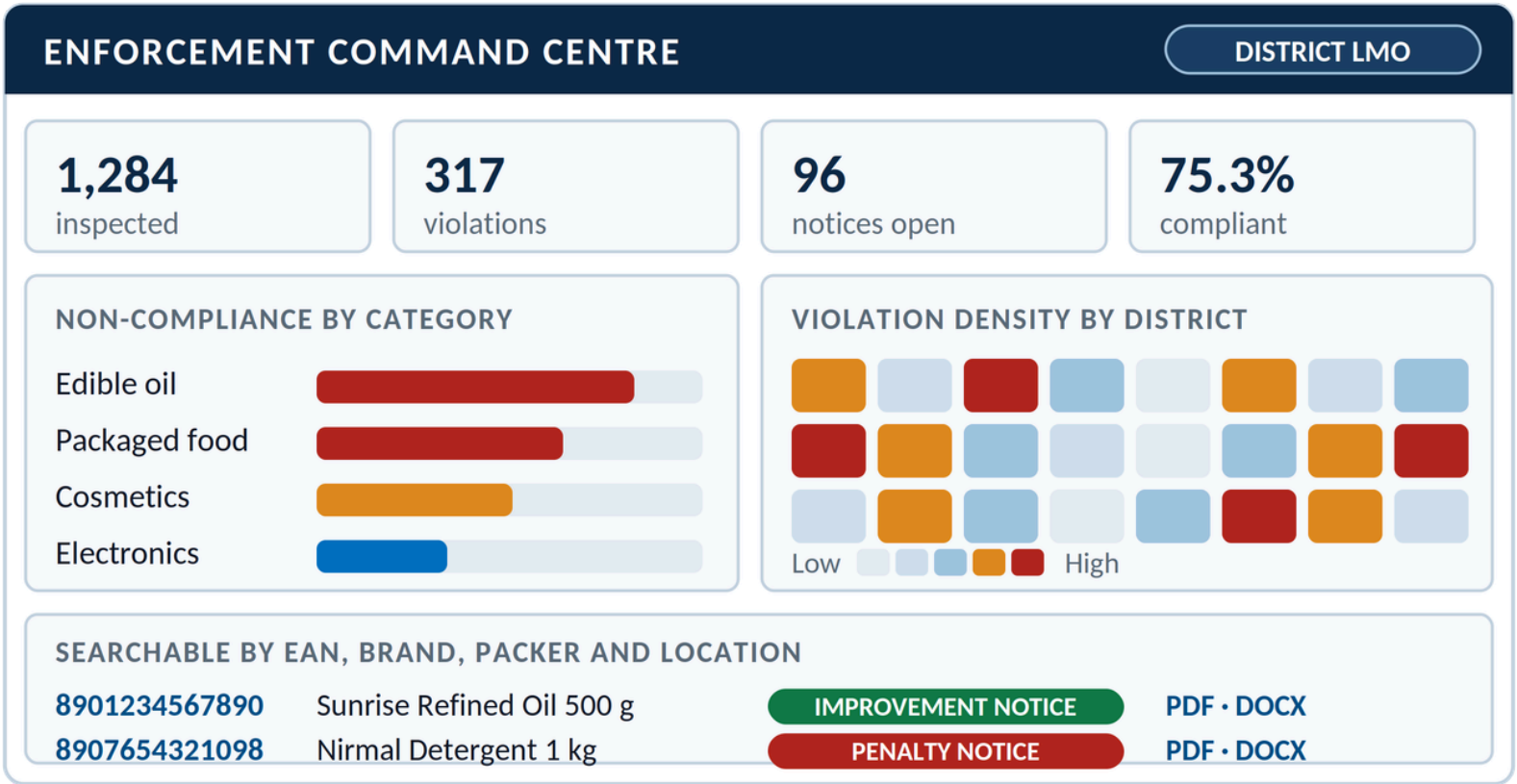
- MRP legibility becomes enforceable rather than advisory
- Recomputed unit price exposes shrinkflation
- Country of origin becomes searchable on e-commerce

HOW A FINDING IS DEALT WITH IN LAW



A camera can measure a label. It cannot weigh a pack — so it never tries.

WHAT THE DEPARTMENT GETS BACK



LEGAL AND REGULATORY FOUNDATION

Legal Metrology Act, 2009 — ss. 2(ea), 18, 36, 48, 50, 52, as amended by the Jan Vishwas Act, 2026
LM (Packaged Commodities) Rules, 2011 — G.S.R. 202(E); rr. 3, 6, 7, 26, 27 and the Fifth Schedule

THE GAZETTE TRAIL WE APPLY

- 23 Jun 2017 · G.S.R. 629(E)
Table I substituted — thresholds move to panel area; Table II omitted
- 28 Mar 2022 · G.S.R. 226(E)
Rule 6(11) — unit sale price, to two decimals
- 6 Oct 2023 · Amendment Rules, 2023
Rule 26(g) for loose e-commerce goods; Fourth Schedule entry 11
- 13 Feb 2026 · G.S.R. 128(E)
Rule 6(10A) — sortable country-of-origin filter, from 1 Jul 2026
- 27 Apr 2026 · G.S.R. 312(E)
Rule 6(10A) substituted and widened, from 1 Jul 2027
- 1 May 2026 · Jan Vishwas Act, 2026
Improvement Notice mechanism comes into force
- 29 May 2026 · G.S.R. 418(E)
Third Amendment — Rule 27 director-level accountability

HOW EACH CLAIM GETS TESTED

Experiment	Method	Target
Accuracy gate	Known point sizes photographed at 15–45°	MAE ≤ 0.3 mm
Caliper corpus	300 retail packs against a digital vernier caliper	MAE ≤ 0.25 mm
Human baseline	30 testers judge 50 borderline packs by eye	Human error at ± 0.2 mm
E-commerce audit	Crawler exercises the origin filter on live listings	Baseline non-compliance rate
Field consultation	Session with a District Legal Metrology Officer	Notice format agreed

STANDARDS AND SOURCES

ISO/IEC 17025:2017, Cl. 7.8.6 — the decision rule must be stated
ILAC G8:09/2019 — guard-banded conformity decisions

- consumeraffairs.gov.in/acts-and-rules · bare Acts and Rules
- indiacode.nic.in · the Act as amended
- egazette.gov.in · every notification cited
- emaap.gov.in · National Legal Metrology Portal